Robust Moving Horizon Estimation with Huber Loss in Presence of Outliers

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***Abstract*—This work presents a robust state estimation strategy based on the Moving Horizon Estimation (MHE) framework, incorporating the Huber loss function (denoted as MHE-H) to improve robustness in the presence of outliers. Traditional MHE and Extended Kalman Filter (EKF) techniques often degrade in accuracy when measurements are affected by large deviations. To address this limitation, the MHE-H method replaces the quadratic penalty on measurement residuals with the Huber loss, which preserves sensitivity to small errors while limiting the impact of outliers. The approach is validated through two case studies: a nonlinear simulation example and an experimental dataset obtained from GNSS/IMU measurements with artificially injected outliers. Results demonstrate that MHE-H significantly reduces the estimation error compared to EKF and standard MHE, while maintaining an acceptable computational cost for real-time applications, making MHE-H a promising alternative for robust estimation in uncertain and noisy environments.**

***Index Terms*—moving horizon estimation, state estimation, outliers, Huber loss**

I. Introduction

State estimation, a fundamental problem in control engineering, is essential for ensuring feedback control, monitoring, and optimization of dynamic systems. Among the various approaches to state estimation, model-based methods have been widely adopted, relying on system dynamics and available sensor measurements to infer the true state of the system. These methods, including Kalman Filtering (KF) [1], particle filters [2], and Moving Horizon Estimation (MHE) [3], leverage mathematical models to integrate noisy observations and provide accurate estimates.

In practice, measurements are often affected by various sources of error, including sensor malfunctions, incomplete or indirect measurements, nonlinearities, model uncertainties, and external disturbances [4]. These factors can introduce biases, degrade estimation accuracy, and, in extreme cases, lead to estimate instability and divergence from true state. In particular, large deviations, commonly referred to as outliers, can significantly impact classical estimation techniques, as in Kalman filter approaches, which typically assume Gaussian noise distributions. Addressing these challenges requires robust estimation methods capable of mitigating the influence of large measurement errors while maintaining reliable state reconstruction.

MHE provides better management of constraints, disturbances, and system nonlinearities compared to Kalman-based methods, making it a strong choice for advanced estimation problems [3]. To obtain a tractable approximation, the MHE method considers a fixed-sized window of the most recent measurements to solve a constrained optimization problem for state estimation, and therefore the horizon window moves ahead in time with the data [5] .

When accurate location estimation is required, one of the challenges is the presence of outliers in position estimates obtained from Global Navigation Satellite Systems (GNSS), which can significantly degrade the overall accuracy of the location estimate [6]. Therefore, robust MHE strategies have been proposed to mitigate the effect of these outliers. A representative approach is the leave-one-out MHE strategy proposed in [7]–[9], which consists of minimizing a series of least-squares problems, each excluding one potentially contaminated measurement, thus achieving robustness at the cost of high computational complexity. Another approach [10] reinterprets MHE from a Bayesian perspective, integrating a robust divergence measure to reduce the impact of outliers rather than merely discarding contaminated measurements.

Robust MHE strategies incorporating the Huber loss [11] have been previously explored. For instance, in [12], robustness against occasional outliers was investigated by applying a Huber-penalized cost function within the MHE framework, focusing on linear systems and comparing the results with standard least-squares MHE. Another example is provided in [13], where a Huber-based MHE method was applied for robust estimation of the position and orientation of moving objects, specifically for tracking tethered airplanes, and was

compared against an ℓ_2-norm MHE formulation and a traditional Extended Kalman Filter.

Building on these previous contributions, we adopt a robust MHE formulation, denoted as MHE-H, which incorporates the Huber loss to reduce the impact of outliers on the measurement error. In contrast to prior works, our approach focuses on a nonlinear simulation example and an experimental case based on real-world GNSS data with artificially injected outliers, highlighting both estimation accuracy and computational efficiency.

This paper is organized as follows: Section II presents the MHE formulation problem. Section III provides a brief description of the Huber loss and describes the robust MHE-H method. Section IV reports simulation results for a nonlinear system with outliers. Section V describes an experimental study using GNSS data with artificially injected outliers to assess the method's performance on real-world measurements. Finally, Section VI presents the conclusions and proposes future works.

NOTATIONS

$\mathbb{N}$ and $\mathbb{R}$ denote the sets of natural and real numbers, respectively; $\mathbb{N}_0$ denotes the positive real numbers; $\|x\|_A^2$ is the squared l_2 norm weighted by A and is calculated as $x^T A x$; $\mathbf{0}_{n\times k}$ denotes an $n \times k$ matrix with all entries of zeros. Given a sequence of vectors $\mathbf{x}_i, \mathbf{x}_{i+1}, \cdots, \mathbf{x}_j$ for $i < j$, we define $\mathbf{x}_{i:j} := (\mathbf{x}_i, \mathbf{x}_{i+1}, \cdots, \mathbf{x}_j)$. The identity matrix of shape $n \times n$ is denoted by $\mathbf{I}_n$.

II. MOVING HORIZON ESTIMATOR

Following [14], we consider the (disturbed) discrete time dynamic system:

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k, \tag{1}$$

$$\mathbf{y}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{v}_k, \tag{2}$$

where $k \in \mathbb{N}_0$ is the current time-step, the subscript $(\cdot)_k$ indicates the time index, $\mathbf{x} \in X \subseteq \mathbb{R}^n$ is the state vector, $\mathbf{u} \subseteq \mathbb{R}^p$ is the vector of control inputs, $\mathbf{y} \subseteq \mathbb{R}^m$ is the measurement vector, $\mathbf{w} \in W \subseteq \mathbb{R}^n$ and $\mathbf{v} \in V \subseteq \mathbb{R}^m$ are the system and measurement noise vectors, respectively. Sets X , W and V are considered closed with $\mathbf{0} \in X$, $\mathbf{0} \in W$ and $\mathbf{0} \in V$. $\mathbf{f} : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^n$ and $\mathbf{h} : \mathbb{R}^n \to \mathbb{R}^m$ are the state and measurement functions, respectively.

A full information estimation approach solves a least squares optimization problem over the history of states by solving the following optimization problem:

$$\min_{\substack{\hat{\mathbf{x}}_{0:k},\\ \hat{\mathbf{w}}_{1:k},\\ \hat{\mathbf{v}}_{1:k}}} \|\hat{\mathbf{x}}_0 - \bar{\mathbf{x}}_0\|^2_{\mathbf{P}_0^{-1}} + \sum_{i=1}^{k} \{\|\hat{\mathbf{w}}_i\|^2_{\mathbf{Q}^{-1}} + \|\hat{\mathbf{v}}_i\|^2_{\mathbf{R}^{-1}}\}$$
$$\text{s. t.} \begin{cases} \hat{\mathbf{x}}_{i+1} = \mathbf{f}(\hat{\mathbf{x}}_i, \mathbf{u}_i) + \hat{\mathbf{w}}_i, \\ \mathbf{y}_i = \mathbf{h}(\hat{\mathbf{x}}_i) + \hat{\mathbf{v}}_i, \\ \hat{\mathbf{x}}_{0:k} \in X, \hat{\mathbf{w}}_{1:k} \in W, \\ \hat{\mathbf{v}}_{1:k} \in V, \end{cases} \tag{3}$$

where $(\hat{\cdot})$ denotes the estimated states of the system (optimization variables), the pair ($\bar{\mathbf{x}}_0 \in \mathbb{R}^n$, $\mathbf{P}_0 \in \mathbb{R}^{n\times n}$) summarizes the prior information at time $k = 0$ where $\bar{\mathbf{x}}_0$ is the current knowledge of the initial estimate and $\mathbf{P}_0$ is the covariance matrix associated with the first initial estimate $\hat{\mathbf{x}}_0$, $\mathbf{Q} \in \mathbb{R}^{n\times n}$ and $\mathbf{R} \in \mathbb{R}^{m\times m}$ are the positive definite process and measurement noise covariance matrices (and their inverses are the information matrices), respectively.

Solving the optimization problem stated in (3) generally becomes intractable for long-term operation due to the increasing number of optimization variables. Moreover, the influence of older states on the current states decreases over time. As an alternative to the full information approach, the MHE strategy can be adopted. To this end, we define the estimation horizon length N for the following finite horizon optimization problem:

$$\min_{\substack{\hat{\mathbf{x}}_{k-N:k},\\ \hat{\mathbf{w}}_{k-N+1:k},\\ \hat{\mathbf{v}}_{k-N+1:k}}} \|\hat{\mathbf{x}}_{k-N} - \bar{\mathbf{x}}_{k-N}\|^2_{\mathbf{P}_{k-N}^{-1}} + \sum_{i=k-N+1}^{k} \{\|\hat{\mathbf{w}}_i\|^2_{\mathbf{Q}^{-1}} + \|\hat{\mathbf{v}}_i\|^2_{\mathbf{R}^{-1}}\}$$
$$\text{s. t.} \begin{cases} \hat{\mathbf{x}}_{i+1} = \mathbf{f}(\hat{\mathbf{x}}_i, \mathbf{u}_i) + \hat{\mathbf{w}}_i, \\ \mathbf{y}_i = \mathbf{h}(\hat{\mathbf{x}}_i) + \hat{\mathbf{v}}_i, \\ \hat{\mathbf{x}}_{k-N:k} \in X, \hat{\mathbf{w}}_{k-N+1:k} \in W, \\ \hat{\mathbf{v}}_{k-N+1:k} \in V, \end{cases} \tag{4}$$

where N is the estimation horizon length, and the first term of the objective function is the arrival cost of the moving horizon problem, which accounts for the prior knowledge of the initial state at time $k - N$ and its associated uncertainty, represented by the covariance matrix $\mathbf{P}_{k-N}$. The optimization problem aims to minimize the difference between expected and measured outputs over a moving horizon window [15].

The model given by (4) achieves good performance when $\mathbf{w}_k$ and $\mathbf{v}_k$ follow a Gaussian distribution. However, the quadratic cost is sensitive to large outliers, which can negatively impact estimation accuracy. To enhance robustness against outliers, we implement a robust MHE in Section III.

III. MHE WITH HUBER PENALTY FUNCTION

The Huber loss function is given by [11]

$$L_\delta(a) = \begin{cases} \|a\|_2^2 & \text{if } \|a\|_2 \le \delta, \\ 2\delta\|a\|_2 - \delta^2 & \text{if } \|a\|_2 > \delta. \end{cases} \tag{5}$$

This loss function is a piecewise function of square loss and absolute loss. This function uses the parameter δ as a threshold to determine whether an outlier is present [16]. Fig. 1 compares the Huber loss function $L_\delta(a)$ (with $\delta = 1$), with the squared loss function and the absolute loss function. As seen in the figure, the Huber loss function behaves quadratically for residuals within the threshold $|a|_2 \le \delta$, similarly to the squared loss, but switches to a linear growth for larger residuals, mimicking the absolute loss. This property makes it more robust to outliers than the purely quadratic loss while preserving sensitivity for small errors.

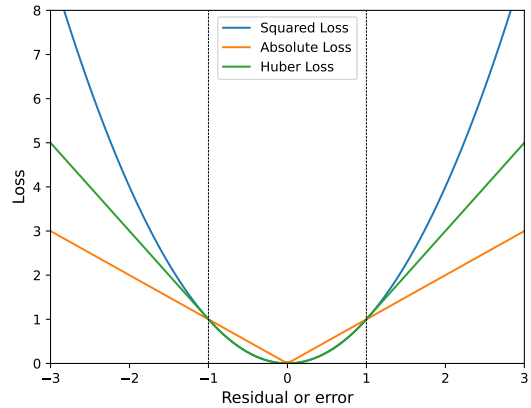


Fig. 1: Comparison of three loss functions.

To handle outliers in $\mathbf{v}_k$, we replace the quadratic cost in the MHE problem (4) with an Huber cost, leading to the MHE-H method. The resulting optimization problem is formulated as:

$$\begin{aligned} \min_{\substack{\hat{\mathbf{x}}_{k-N:k},\\ \hat{\mathbf{w}}_{k-N+1:k},\\ \hat{\mathbf{v}}_{k-N+1:k}}} & \|\hat{\mathbf{x}}_{k-N} - \bar{\mathbf{x}}_{k-N}\|^2_{\mathbf{P}^{-1}_{k-N}} + \\ & \sum_{i=k-N+1}^{k} \{\|\hat{\mathbf{w}}_i\|^2_{\mathbf{Q}^{-1}} + L_\delta(\hat{\mathbf{v}}_i)\} \\ \text{s. t.} & \begin{cases} \hat{\mathbf{x}}_i = \mathbf{f}(\hat{\mathbf{x}}_{i+1}, \mathbf{u}_i) + \hat{\mathbf{w}}_i, \\ \mathbf{y}_i = \mathbf{h}(\hat{\mathbf{x}}_i) + \hat{\mathbf{v}}_i, \\ \hat{\mathbf{x}}_{k-N:k} \in X, \hat{\mathbf{w}}_{k-N+1:k} \in W, \\ \hat{\mathbf{v}}_{k-N+1:k} \in V. \end{cases} \end{aligned} \tag{6}$$

Measurement errors within the limit δ use the quadratic loss, and outside this limit they use the absolute loss, so large measurement errors are penalized less severely, which increases the robustness of the model against outliers [17]. Finding the optimal δ value may require experimentation or cross-validation. Alternatively, the tuning parameter can be simultaneously optimized to improve the estimator performance.

IV. Simulation Example

For the simulation test, we considered the following discrete-time nonlinear system [18]:

$$\begin{aligned} x_{k+1}(1) &= 0.99x_k(1) + 0.2x_k(2), \\ x_{k+1}(2) &= -0.1x_k(1) + \frac{0.5x_k(2)}{1 + x_k^2(2)} + w_k, \\ y_k &= x_k(1) - 3x_k(2) + v_k, \end{aligned} \tag{7}$$

where the noise characteristics are:

$$\begin{aligned} v_k &\sim \varrho\, \mathcal{N}(0, \sigma_v^2) + (1 - \varrho)\, \mathcal{N}(0, \sigma_o^2), \\ w_k &= |\xi_k|, \text{ with } \xi_k \sim \mathcal{N}(0, \sigma_w^2). \end{aligned} \tag{8}$$

This implies that v_k is generated from a Gaussian distribution with variance σ_v^2 with probability ϱ, and from another Gaussian distribution with a larger variance σ_o^2 (representing outliers) with probability $1 - \varrho$. Unlike v_k, which is symmetrically distributed around zero, w_k is generated by taking the absolute value of a zero-mean Gaussian distribution with covariance σ_w^2. In this case, the variances σ_v^2, σ_w^2, σ_o^2 were 0.01, 1, and 100, respectively. The probability ϱ was set to 0.9, which implies that approximately 10% of the measurements were outliers. The initial state of the system was $\mathbf{x}_0 = [1, 0]^T$.

The implemented MHE-H algorithm was compared with an Extended Kalman Filter (EKF) and a traditional MHE estimator. For the three estimation solvers, we set the weighting parameters to $Q = 1$, $R = 0.01$ and $\mathbf{P}_0 = \mathbf{I}_2$ and we assumed the initial state to be $\bar{\mathbf{x}}_0 = [0, 0]^T$. For the moving horizon estimator approaches, we set the horizon length to $N = 10$. The threshold parameter $\delta = 0.1$ for the Huber loss in (5) was chosen to ensure that the nominal measurement noise lies within the quadratic region of the Huber function, while outlier values fall into its linear region.

The performance metric utilized to evaluate estimation accuracy was the mean squared error (MSE):

$$MSE = \frac{1}{Kn} \sum_{k=1}^{M} \sum_{i=0}^{n} (\mathbf{x}_k(i) - \hat{\mathbf{x}}_k(i))^2 \tag{9}$$

where K is the number of measurements and n is the length of the state vector. The MSE presented in the tables were obtained by averaging over 100 independent simulation runs.

In addition to estimation accuracy, the mean computational time (MCT, in milliseconds) was also used as a performance metric. The MCT is defined as the average processing time per time step required to produce an estimate, and reflects the computational efficiency of the algorithm.

To compute the arrival cost in MHE and MHE-H, the prior state $\bar{\mathbf{x}}_{k-N}$ and the associated covariance matrix $\mathbf{P}_{k-N}$ are recursively updated using an Extended Kalman Filter (EKF).

Both MHE and MHE-H estimators were implemented in Python using the library CasADi: an open-source symbolic framework used to perform numeric optimization [19]. Then, the optimization problem is solved using an interior point method (IPOPT) [20]. All simulations were performed on desktop computer equipped with a 2.8 GHz Intel i5 with 16 GB RAM, running the 64-bit version of Ubuntu Linux 22.04.

A side-by-side comparison of the estimators from one simulation run is presented in Fig. 2. The plots illustrate the estimated trajectories for the two state variables (Figs. 2a and 2b) as well as the corresponding estimation squared errors (Figs. 2c and 2d). The MHE-H demonstrates its effectiveness against outliers through time by consistently producing the lowest error values, particularly noticeable in the squared error plots, where its values remain lower compared to those of both EKF and traditional MHE.

This visual evidence is consistent with the quantitative results reported in Table I. As expected, the EKF is the most computationally efficient method, with the lowest MCT value, but it also yields the highest MSE, indicating poorer estimation accuracy under outlier contamination. The MHE-H approach achieves the lowest MSE, about 75% lower than the EKF and 65% lower than the traditional MHE.

The improved performance of MHE-H over the EKF can be attributed to the fact that the EKF relies on the assumption of Gaussian process and measurement noise [21]. When this assumption is violated, as in the case of occasional outliers, the EKF tends to perform suboptimally, whereas the robust formulation of MHE-H handles such deviations more effectively.

TABLE I: Simulation example: Average MSE and MCT (in ms) computed over 100 Monte Carlo runs.

Estimator	EKF	MHE	MHE-H
MSE	1.01	0.756	0.258
MCT	0.221	3.71	45.6

The high computational cost of the MHE-H approach is largely due to the fact that, unlike the EKF, it solves a constrained nonlinear optimization problem at each time step. This process involves repeatedly evaluating the system dynamics, measurement functions, and their derivatives, which makes it significantly more expensive than the recursive equations of the EKF. Compared to the standard MHE, the MHE-H formulation increases the computational burden by replacing the purely quadratic cost with the Huber loss, which is a piecewise-defined non-quadratic cost function, requiring the solver to handle potential non-differentiabilities and irregular curvature in the objective. Despite this, it still benefits from the moving horizon strategy, which significantly reduces the computational cost compared to the full information problem stated in (3). Given the current MCT of the MHE-H, the method is suitable for real-time applications with sampling frequencies up to approximately 20 Hz.

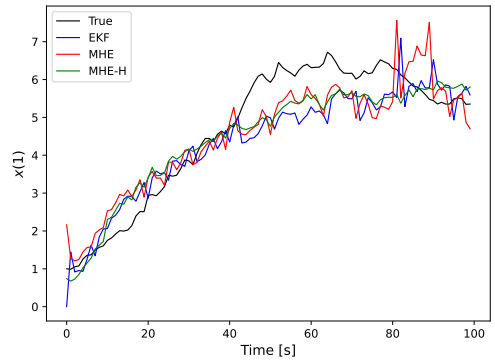
V. Results with Experimental Data

To evaluate the robustness of our method in a real-world setting, we used data from a series of experiments performed with a small-scale ground robot developed in-house. The platform was equipped with an ArduSimple SimpleRTK2b module, based on the Ublox ZED-F9P GNSS receiver, configured to provide both position and velocity measurements. An Xsens MTi-630 IMU was also integrated to provide acceleration measurements used in the model. The dataset considered in this work corresponds to an experiment in which the vehicle was teleoperated to follow a square trajectory aligned with the cardinal directions, starting toward the east and executing counterclockwise turns. In a preprocessing stage, the position measurements given by the GNSS reciever were converted from the geocentric Earth-Centered, Earth-Fixed (ECEF) Cartesian coordinates to the local East-North-Up (ENU) Cartesian coordinates [22]. The measurements of the position after the coordinate frame transformation are shown in Fig. 3.

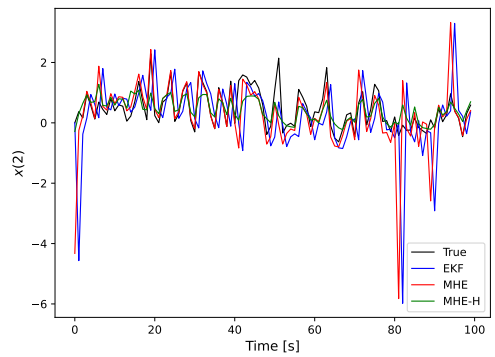
The discrete-time kinematic model used in this experiment is:

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k + \mathbf{w}_k, \\ \mathbf{y}_k &= \mathbf{C}\mathbf{x}_k + \mathbf{v}_k, \end{aligned} \tag{10}$$

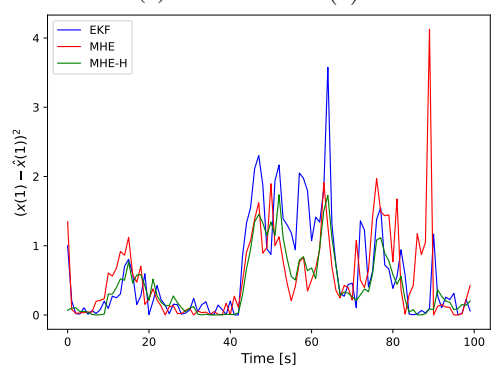
where the state vector is $\mathbf{x} = [p_x, p_y, \dot{p}_x, \dot{p}_y]$, with p_x, p_y denoting the position coordinates and $\dot{p}_x, \dot{p}_y$ the velocities. The



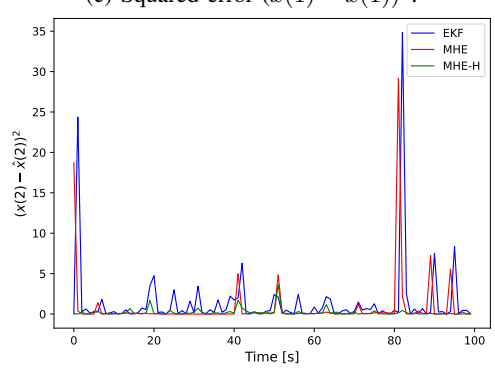
(a) Estimation of $x(1)$.



(b) Estimation of $x(2)$.



(c) Squared error $(x(1) - \hat{x}(1))^2$.



(d) Squared error $(x(2) - \hat{x}(2))^2$.

Fig. 2: Comparison of estimators for simulation example.

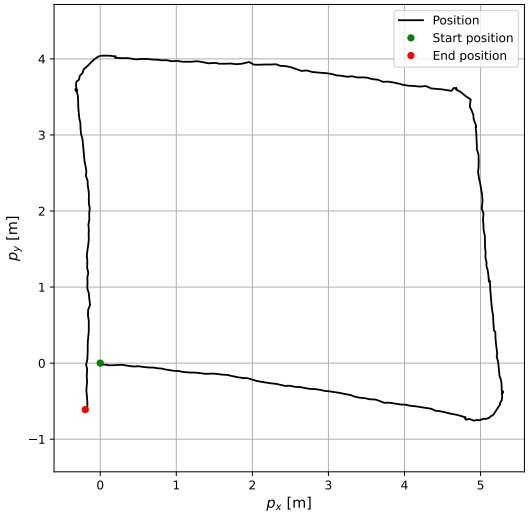


Fig. 3: Evolution of the position.

input vector is $\mathbf{u} = [a_x, a_y]$ where a_x, a_y are the accelerations measured by the IMU. The matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ are:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \frac{1}{2}\Delta t^2 & 0 \\ 0 & \frac{1}{2}\Delta t^2 \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

The GNSS sensor operated at a sampling rate of 5 Hz, while the IMU operated at 20 Hz. Consequently, a sampling interval of $\Delta t = 0.2$ s was adopted in the model. Due to the different sampling rates, IMU measurements were temporally aligned with the GNSS timestamps by matching each GNSS sample to the closest IMU sample.

To assess robustness, GNSS position measurements were post-processed by injecting artificial outliers, while the original GNSS data served as ground-truth states for evaluation. We model the artificial measurement noise as $\mathbf{v}_k \sim \varrho\mathcal{N}\left(\mathbf{0}, \sigma_o^2\mathbf{I}_2\right)$ with an outlier occurrence probability of $\varrho = 0.1$ and deviation $\sigma_o = 1.5$ m. Although the dataset was acquired in Real-Time Kinematic (RTK) mode, which provides centimeter-level accuracy, the value $\sigma_o = 1.5$ m was chosen to emulate the typical horizontal position error of the receiver operating in standalone Position–Velocity–Time (PVT) mode, as reported in the Ublox ZED-F9P datasheet. This choice reflects occasional drops from RTK to PVT performance and sets a realistic scale for injected outliers.

Just like in the previous example, the implemented MHE-H algorithm was compared with an EKF and a traditional MHE estimator. For the three estimation solvers, we set the weighting parameters to $Q = 10\mathbf{I}_4$, $R = 0.1\mathbf{I}_2$ and $\mathbf{P}_0 = \mathbf{I}_2$ and we assumed the initial state to be $\bar{\mathbf{x}}_0 = [0, 0, 0, 0]^T$. For the moving horizon estimator approaches, we set the horizon length to $N = 10$. The threshold parameter for the Huber loss (5) was tuned to $\delta = 0.05$.

Fig. 4 illustrates the estimated trajectories of position and velocity, together with the artificially corrupted measurements and the original GNSS values used as reference. It can be observed that while EKF and MHE are strongly affected by the injected outliers, the proposed MHE-H provides smoother and more accurate estimates. Table II complements this analysis by reporting quantitative results in terms of mean and maximum estimation errors, as well as the MCT. The maximum position error of the MHE-H is about 70% lower than the maximum position error of the EKF and the MHE. The maximum velocity error of the MHE-H is about 35% lower than the maximum velocity error of the EKF and the MHE. Even though the MCT of MHE-H is higher than that of the EKF as expected, the value of 5.61 ms per step remains well below the sampling period of the GNSS sensor, ensuring real-time feasibility. It is also worth noting that this value is significantly lower than the MCT obtained in the nonlinear simulation example, which can be attributed to the simpler structure of the linear kinematic model used in the experimental setup. These results indicate that MHE-H achieves a significant reduction in position and velocity estimation errors while maintaining a computational cost compatible with online implementation.

TABLE II: Average and maximum position and velocity estimation error and MCT for experimental data example.

Estimator	EKF	MHE	MHE-H
Mean position error [m]	0.175	0.127	0.0954
Max position error [m]	4.17	4.09	1.31
Mean velocity error [m/s]	0.605	0.576	0.534
Max velocity error [m/s]	3.28	3.50	2.18
Mean computation time [ms]	0.329	2.96	5.61

VI. Conclusion

This work presented a robust Moving Horizon Estimation method based on the Huber loss function (MHE-H) for state estimation in systems affected by outliers. The MHE-H approach was compared against a standard EKF and a traditional MHE in two case studies: a nonlinear simulation example and an experimental dataset obtained from GNSS/IMU measurements with artificially injected outliers. In both scenarios, MHE-H consistently outperformed EKF and MHE in terms of estimation accuracy, achieving reductions of more than 75% in mean squared error in simulation and up to 70% in maximum position error in the experimental test. Despite its higher computational cost, MHE-H remains feasible for real-time applications operating at moderate sampling rates. Overall, MHE-H provides a robust and effective alternative to classical estimators, particularly under outlier conditions. Future work includes the integration of faster optimization solvers to reduce computational cost, as well as the implementation and evaluation of the MHE-H approach in embedded real-time localization systems.

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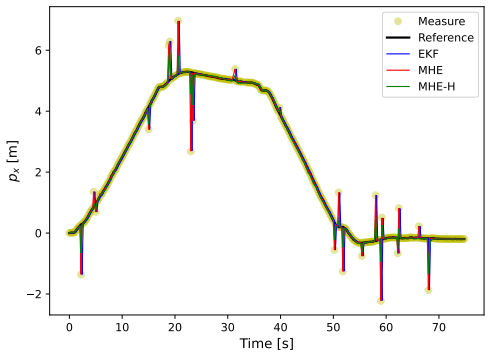
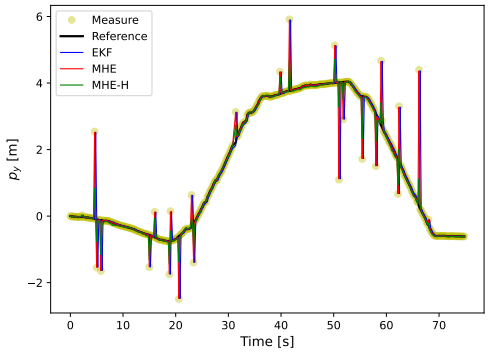
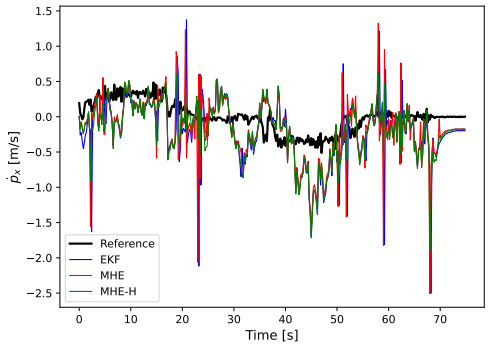


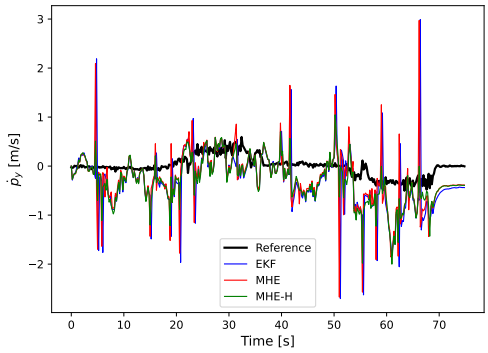
(a) Estimation of p_x.



(b) Estimation of p_y.



(c) Estimation of $\dot{p}_x$.



(d) Estimation of $\dot{p}_y$.

Fig. 4: Comparison of estimators for real data example.

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